Density Matrices and Quantum Communication at Nonzero Spin–Orbit Coupling

Conditional spin and helicity-pseudospin states in the Hubbard–Rashba BdG model

Technical derivation and validated numerical demonstration

November 12, 2026

Abstract

This report explains how to construct a two-qubit density matrix when the Rashba coupling V_{SO} is nonzero. The central change from the zero-SOC Oh–Kim construction is that scalar normal and anomalous correlations are no longer sufficient. One must retain 2×2 correlator matrices, transform the helicity pairing with the correct momentum-dependent spinors and gap phase, apply Wick's theorem, and only then normalize the sector containing one detected electron at each of two positions. The resulting original-spin state is generally neither an X state nor a Werner state. A helicity-band calculation is also possible, but it describes helicity-pseudospin qubits whose local modes and operations are not automatically physical spin qubits and spin operations. We derive general concurrence, fixed-reference and optimized average teleportation fidelities, and the equal-probability Pauli-ensemble Holevo quantity used for dense coding. A separately archived 101×101 demonstration at $V_{\mathrm{SO}}/t = 0$ and 0.5 validates the construction: the zero-SOC Oh–Kim matrix is recovered to 5.55×10^{-17}, all 24 conditional matrices pass normalization and positivity checks, and the dense-coding implementation agrees with the repository's independent Holevo routine to 2.22×10^{-16}. The nonzero-SOC example is a methodological demonstration, not a converged SOC-dependence study or a claim of experimental helicity Bell measurement.

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1 Executive overview

At $V_{\mathrm{SO}} = 0$, the two helicity branches are degenerate and the original spin basis supports the Oh–Kim assumptions: spin-diagonal normal propagation and spin-singlet anomalous propagation. These assumptions lead to an X state and, for the conditional two-spin reduction used here, a Werner state.

At $V_{\mathrm{SO}} \neq 0$, spin and momentum are locked. The original-spin normal correlator can acquire off-diagonal components, while the pair correlator contains mixed singlet–triplet structure. Consequently, the correct generalization is not a new scalar $g(\mathbf{r}), f(\mathbf{r})$ formula. It is the matrix contraction

$$\boxed{\widetilde{\rho}_{\alpha\beta,\alpha'\beta'} = N_{B,\beta'\beta} N_{A,\alpha'\alpha} - N_{BA,\beta'\alpha} N_{AB,\alpha'\beta} + F^{*}_{AB,\alpha'\beta'} F_{AB,\alpha\beta}} \tag{1}$$

followed by $\rho_{AB} = \widetilde{\rho}_{AB} / \operatorname{Tr} \widetilde{\rho}_{AB}$. The three terms are the direct, normal-exchange, and anomalous-pair contributions.

The following distinction governs every quantum-information interpretation in this report.

Choice	Internal qubit states	Operational meaning
Original spin	$\uparrow, \downarrow$ at A, B	Ordinary spin entanglement and spin protocols; generally a non-X, non-Werner state at nonzero SOC.
Helicity band	$s = +1, -1$ at band-filtered modes	Band-pseudospin entanglement and protocols; requires band-pseudospin local unitaries, Bell measurement, and feed-forward.

The demonstration finds ideal singlets at zero separation, but no concurrence, teleportation advantage, or dense-coding advantage at the tested nonzero lattice vectors for either basis. This is a limited numerical observation at one mesh and parameter set, not a general theorem about Rashba superconductors.

2 Model, symmetries, and the meaning of helicity

The square-lattice normal-state dispersion is

$$\epsilon_{\mathbf{k}s} = -2t(\cos k_x + \cos k_y) - 2sV_{\mathrm{SO}}\lambda_{\mathbf{k}}, \qquad \lambda_{\mathbf{k}} = \sqrt{\sin^2 k_x + \sin^2 k_y}, \tag{2}$$

with $s = \pm 1$. The BdG quasiparticle energy and zero-temperature occupation are

$$E_{\mathbf{k}s} = \sqrt{(\epsilon_{\mathbf{k}s} - \mu)^2 + |\Delta_{\mathbf{k}s}|^2}, \qquad v^2_{\mathbf{k}s} = \frac{1}{2}\left(1 - \frac{\epsilon_{\mathbf{k}s} - \mu}{E_{\mathbf{k}s}}\right). \tag{3}$$

The self-consistency equations determine Δ_0, Δ_S, μ anew for each SOC and filling. A converged stationary solution is required before any density matrix is computed; a residual check alone, however, does not establish global thermodynamic stability.

2.1 What changes when V_{SO} is turned on?

At zero SOC, spin rotational symmetry makes the original spin basis natural. At nonzero SOC, the normal Hamiltonian is diagonal in a helicity basis whose spin quantization axis depends on $\mathbf{k}$. Helicity is therefore a useful band quantum number, but it is not a single global spin projection shared by all momenta.

This has three distinct consequences:

(i) original-spin normal correlations need not be diagonal;

(ii) time-reversed intraband pairing transforms into mixed original-spin singlet–triplet correlations;

(iii) an on-site "helicity qubit" is not unique unless the band-filtered modes are explicitly defined.

None of these statements makes original-spin entanglement undefined. Entanglement is a property of a state *and a tensor-product decomposition*. One may choose original-spin modes or helicity-band modes, but the two choices define different resources.

3 Helicity transformation and full pairing phase

Hutchinson, Hirsch, and Marsiglio define [1]

$$c^\dagger_{\mathbf{k}s} = \frac{c^\dagger_{\mathbf{k}\uparrow} + s e^{i\theta_{\mathbf{k}}} c^\dagger_{\mathbf{k}\downarrow}}{\sqrt{2}}, \qquad e^{i\theta_{\mathbf{k}}} = \frac{\sin k_y - i \sin k_x}{\lambda_{\mathbf{k}}}. \tag{4}$$

Inverting the transformation gives $c_{\mathbf{k}\sigma} = \sum_s U_{\mathbf{k},\sigma s} c_{\mathbf{k}s}$ with

$$U_{\mathbf{k}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ e^{i\theta_{\mathbf{k}}} & -e^{i\theta_{\mathbf{k}}} \end{pmatrix}, \qquad U_{-\mathbf{k}} = Z U_{\mathbf{k}}, \quad Z = \mathrm{diag}(1,-1), \tag{5}$$

because $e^{i\theta_{-\mathbf{k}}} = -e^{i\theta_{\mathbf{k}}}$.

The mean field pairs time-reversed electrons within the same helicity band. Its full gap is

$$\boxed{\Delta_{\mathbf{k}s} = s e^{-i\theta_{\mathbf{k}}} \bar{\Delta}_{\mathbf{k}s}.} \tag{6}$$

The repository's `bdg["Delta_ks"]` stores the even scalar $\bar{\Delta}_{\mathbf{k}s}$. Omitting the phase in Eq. (6) destroys the correct relation between $\mathbf{k}$ and $-\mathbf{k}$ and can produce spurious same-band pairing.

At $T = 0$, the pair mean field satisfies $b_{\mathbf{k}s} = \bar{\Delta}_{\mathbf{k}s}/(2E_{\mathbf{k}s})$. Up to one common sign that cancels from F^*F, define

$$N^{\mathrm{h}}_{\mathbf{k}} = \begin{pmatrix} v^2_{\mathbf{k}+} & 0 \\ 0 & v^2_{\mathbf{k}-} \end{pmatrix}, \tag{7}$$

$$F^{\mathrm{h}}_{\mathbf{k}} = \begin{pmatrix} e^{-i\theta_{\mathbf{k}}} \bar{\Delta}_{\mathbf{k}+}/(2E_{\mathbf{k}+}) & 0 \\ 0 & -e^{-i\theta_{\mathbf{k}}} \bar{\Delta}_{\mathbf{k}-}/(2E_{\mathbf{k}-}) \end{pmatrix}. \tag{8}$$

Here $N_{\mathbf{k},ss'} = \langle c^\dagger_{\mathbf{k}s} c_{\mathbf{k}s'} \rangle$ and $F_{\mathbf{k},ss'} = \langle c_{\mathbf{k}s} c_{-\mathbf{k}s'} \rangle$.

3.1 TRIM and Pauli exclusion

At a time-reversal-invariant momentum (TRIM), $\lambda_{\mathbf{k}} = 0$ and the helicity phase is singular. For nondegenerate same-band pairing, $c_{\mathbf{k}s}$ and $c_{-\mathbf{k}s}$ coincide up to a gauge, so the pair amplitude must vanish by Pauli exclusion. The implementation enforces this at TRIM points. At exact zero-SOC degeneracy, the original-spin singlet is instead gauge invariant and is treated separately.

Off a TRIM, the full phase gives

$$F^{\mathrm{h}}_{-\mathbf{k}} = -F^{\mathrm{h}}_{\mathbf{k}}. \tag{9}$$

The numerical maximum off-TRIM residual in Eq. (9) is 3.30×10^{-15}.

4 Matrix-valued correlators

4.1 Transforming back to original spin

For the creation–annihilation convention $N_{\sigma\sigma'} = \langle c^\dagger_\sigma c_{\sigma'}\rangle$, the correct transformations are

$$\boxed{N^{\rm spin}_{\mathbf{k}} = U^*_{\mathbf{k}} N^{\rm h}_{\mathbf{k}} U^T_{\mathbf{k}}, \qquad F^{\rm spin}_{\mathbf{k}} = U_{\mathbf{k}} F^{\rm h}_{\mathbf{k}} U^T_{-\mathbf{k}}.} \tag{10}$$

The first expression is not $UNU^\dagger$ under this index convention. Direct multiplication confirms the phase of $\langle c^\dagger_\uparrow c_\downarrow\rangle$. The anomalous transformation uses both $U_{\mathbf{k}}$ and $U_{-\mathbf{k}}$ because the two electrons have opposite momenta.

At zero SOC, the two bands are degenerate and Eq. (10) restores a scalar normal matrix and a singlet anomalous matrix. At nonzero SOC, the normal matrix can have off-diagonal elements and the anomalous matrix can contain both singlet and triplet structures.

4.2 Fourier transformation to two positions

Let A, B be lattice positions and $\mathbf{r} = \mathbf{R}_B - \mathbf{R}_A$. For either basis,

$$N_{AB}(\mathbf{r}) = \frac{1}{N_k^2}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} N_{\mathbf{k}}, \qquad F_{AB}(\mathbf{r}) = \frac{1}{N_k^2}\sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} F_{\mathbf{k}}. \tag{11}$$

Uniformity gives $N_A = N_B = N_{AB}(0)$ and $N_{BA} = N^\dagger_{AB}$. Direct anticommutation at the same ordered positions gives

$$F_{BA,\beta\alpha} = \langle c_{B\beta} c_{A\alpha}\rangle = -F_{AB,\alpha\beta}. \tag{12}$$

At $A = B$, $F_{AB}(0) = -F^T_{AB}(0)$ exactly. The code removes only the finite-quadrature symmetric residue at this point. It does not repair an unphysical state by clipping eigenvalues.

5 Conditional Wick construction of the density matrix

5.1 Detection sector

The full BCS state contains variable particle number. A two-qubit communication resource is defined only after selecting the sector with one electron detected at A and one at B. This is the same conditional logic as the Oh–Kim reduction, but without assuming scalar spin propagation.

For a Gaussian BdG mean-field state, Wick's theorem gives

$$\begin{aligned}\widetilde{\rho}_{\alpha\beta,\alpha'\beta'} &= \langle c^\dagger_{B\beta'} c^\dagger_{A\alpha'} c_{A\alpha} c_{B\beta}\rangle \\ &= N_{B,\beta'\beta} N_{A,\alpha'\alpha} - N_{BA,\beta'\alpha} N_{AB,\alpha'\beta} + F^*_{AB,\alpha'\beta'} F_{AB,\alpha\beta}.\end{aligned} \tag{13}$$

The basis ordering is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. The normalized matrix and detection-sector weight are

$$P_{AB} = \mathrm{Tr}\,\widetilde{\rho}_{AB}, \qquad \rho_{AB} = \frac{\widetilde{\rho}_{AB}}{P_{AB}}. \tag{14}$$

The weight is retained in the output. It should not be confused with a communication fidelity.

The creation-side anomalous factor is $F^*_{AB,\alpha'\beta'}$, equivalently $-F^*_{BA,\beta'\alpha'}$. This sign is essential: it produces the negative singlet coherence and the exact zero-SOC limit.

5.2 Why the result is a valid state

The unnormalized matrix is a projected positive two-detection operator. Therefore, for a physical Gaussian input state and exact arithmetic,

$$\widetilde{\rho}_{AB} = \widetilde{\rho}_{AB}^{\dagger}, \qquad \widetilde{\rho}_{AB} \geq 0, \qquad P_{AB} > 0. \tag{15}$$

After normalization, ρ_{AB} is Hermitian, positive semidefinite, and has unit trace. These properties are checked numerically rather than assumed from a target X-matrix form.

At $A = B$, Pauli exclusion leaves only the antisymmetric internal state, so the conditional matrix is the same-site singlet. The test recovers it to 1.11×10^{-16} in both bases.

6 Exact recovery of the zero-SOC Oh–Kim state

The Oh–Kim reduction [2] follows when

$$N_A = N_B = n_0 I, \qquad N_{AB} = N_{BA} = n_0 g(\mathbf{r}) I, \qquad F_{AB} = n_0 f(\mathbf{r}) i\sigma_y. \tag{16}$$

Then Eq. (13) reduces to

$$\rho_X = \frac{1}{\mathcal{N}} \begin{pmatrix} 1-g^2 & 0 & 0 & 0 \\ 0 & 1+f^2 & -(g^2+f^2) & 0 \\ 0 & -(g^2+f^2) & 1+f^2 & 0 \\ 0 & 0 & 0 & 1-g^2 \end{pmatrix}, \tag{17}$$

where $g^2 = |g|^2$, $f^2 = |f|^2$, and $\mathcal{N} = 4 - 2g^2 + 2f^2$. Defining

$$w = \frac{g^2 + f^2}{2 + f^2 - g^2}, \qquad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \tag{18}$$

gives the Werner representation

$$\rho_X = (1-w)\frac{I_4}{4} + w|\Psi^-\rangle\langle\Psi^-|. \tag{19}$$

This special structure explains why fixed and optimized teleportation formulas coincide in the earlier $V_{\mathrm{SO}} = 0$ study. It must not be imposed at nonzero SOC.

The implementation recovers Eq. (17) with maximum elementwise error 5.55×10^{-17} in the executed zero-SOC test.

7 Spin qubits versus helicity-pseudospin qubits

7.1 Original-spin modes

For `basis="spin"`, the internal states are the original $\uparrow, \downarrow$ modes at A, B. This is the appropriate construction for ordinary spin entanglement and spin teleportation. Nonzero SOC changes the state but not the definition of these local spin modes.

7.2 Band-lattice modes

For `basis="band"`, the implementation uses

$$d_{i,s} = \frac{1}{N_k} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_i} c_{\mathbf{k}s}. \tag{20}$$

These are canonical fermionic modes and therefore define a valid mathematical qubit basis after conditional detection. They are, however, nonlocal combinations of original spin/site modes. The corresponding band-pseudospin singlet is not generally the original-spin singlet.

Because the helicity transformation depends on momentum, the original-spin and band-lattice states are not related by a fixed pair of local spin rotations. Their concurrences, entropies, and communication benchmarks can differ. This is not a numerical contradiction; it is a change of tensor-product decomposition and operational resource.

7.3 Operational assumptions

For helicity-pseudospin teleportation, one must assume independently addressable A, B band modes, an independently prepared input pseudospin Q, ideal band-basis Bell measurement on Q, A, and ideal classical feedforward to B. For helicity-pseudospin dense coding, one must similarly implement band-pseudospin Pauli encodings, transmit the encoded band mode, and perform joint decoding.

The calculation does not establish microscopic Bell-measurement feasibility, individual Cooper-pair addressability, or an experimental band-filter implementation. For ordinary physical spin protocols, the original-spin basis is the unambiguous choice.

8 Entanglement at nonzero SOC

The generalized state need not be an X state. Concurrence is therefore calculated using Wootters' general two-qubit formula [3]:

$$R = (\sigma_y \otimes \sigma_y)\rho^*_{AB}(\sigma_y \otimes \sigma_y), \tag{21}$$

$$C(\rho_{AB}) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}, \tag{22}$$

where λ_i are the decreasingly ordered square roots of the eigenvalues of $\rho_{AB}R$. In the band calculation, σ_y acts on band pseudospin.

Several quantities must not be conflated:

- a large anomalous correlator F is not itself concurrence;
- a nonzero superconducting gap does not guarantee finite-site entanglement;
- zero concurrence does not imply zero singlet overlap or zero Holevo quantity;
- concurrence in the original-spin basis need not equal concurrence in the band-pseudospin basis.

The zero-separation conditional state has $C = 1$. At every tested nonzero lattice vector in the demonstration, $C = 0$ for both bases. This statement is limited to the reported parameter set and mesh.

9 Teleportation fidelity

9.1 Fixed Bell-reference protocol

Let $F_{\mathrm{h}} = \langle \Psi_{\mathrm{h}}^{-} | \rho_{AB} | \Psi_{\mathrm{h}}^{-} \rangle$ be the overlap with the chosen reference Bell state in the same qubit basis. For the fixed protocol referenced to that state, the average over uniformly distributed pure input qubits is [4, 5]

$$\overline{\mathcal{F}}_{\mathrm{ref}} = \frac{2F_{\mathrm{h}} + 1}{3}. \tag{23}$$

The reference overlap is not automatically the fully entangled fraction.

9.2 Locally optimized protocol

Define the real correlation matrix

$$T_{ij} = \mathrm{Tr}[\rho_{AB}(\sigma_i \otimes \sigma_j)]. \tag{24}$$

The fully entangled fraction, optimized over local single-qubit unitaries, is

$$\mathfrak{f} = \max_{U_A, U_B} \langle \Phi^{+} | (U_A \otimes U_B) \rho_{AB} (U_A \otimes U_B)^{\dagger} | \Phi^{+} \rangle = \frac{1 + \sum_i \nu_i(T)}{4}, \tag{25}$$

where $\nu_i(T)$ are singular values. The optimized average fidelity is

$$\overline{\mathcal{F}}_{\mathrm{opt}} = \frac{2\mathfrak{f} + 1}{3}. \tag{26}$$

Consequently,

$$\overline{\mathcal{F}}_{\mathrm{opt}} > \frac{2}{3} \Longleftrightarrow \mathfrak{f} > \frac{1}{2} \Longleftrightarrow \sum_i \nu_i(T) > 1. \tag{27}$$

Local optimization is performed in the selected spin or band-pseudospin basis. The latter assumes band-pseudospin local unitaries.

For a Werner state of weight w,

$$F_{\mathrm{h}} = \mathfrak{f} = \frac{1 + 3w}{4}, \qquad \overline{\mathcal{F}}_{\mathrm{ref}} = \overline{\mathcal{F}}_{\mathrm{opt}} = \frac{1 + w}{2}. \tag{28}$$

For a general nonzero-SOC state, these identities need not hold.

At $V_{\mathrm{SO}} = 0.5t$ and $\mathbf{r} = (1, 0)$, the original-spin example has

$$F_{\mathrm{h}} = 0.251401965, \quad \mathfrak{f} = 0.251881235, \quad \overline{\mathcal{F}}_{\mathrm{ref}} = 0.500934644, \quad \overline{\mathcal{F}}_{\mathrm{opt}} = 0.501254157. \tag{29}$$

The band example instead has

$$F_{\mathrm{h}} = 0.249609387, \quad \mathfrak{f} = 0.250454072, \quad \overline{\mathcal{F}}_{\mathrm{ref}} = 0.499739591, \quad \overline{\mathcal{F}}_{\mathrm{opt}} = 0.500302714. \tag{30}$$

Fixed and optimized fidelities differ, and all are far below 2/3.

10 Dense coding with SOC

10.1 Protocol and Pauli ensemble

Dense coding uses a shared ρ_{AB}, local encoding on A, transmission of A to B, and joint decoding [6, 7]. The implemented ensemble applies I, X, Y, Z on A with equal probability. In a band calculation these are helicity-pseudospin Pauli operations.

The ensemble average is

$$\bar{\rho}_{AB} = \frac{1}{4}\sum_{i=0}^{3}(\sigma_i \otimes I)\rho_{AB}(\sigma_i \otimes I)^{\dagger}. \tag{31}$$

The Pauli twirl identity gives, for any two-qubit state,

$$\bar{\rho}_{AB} = \frac{I_A}{2} \otimes \rho_B, \qquad \rho_B = \mathrm{Tr}_A\, \rho_{AB}. \tag{32}$$

Therefore the Holevo quantity of this ensemble is

$$\boxed{\chi_{\mathrm{P}} = S_2(\bar{\rho}_{AB}) - S_2(\rho_{AB}) = 1 + S_2(\rho_B) - S_2(\rho_{AB}),} \tag{33}$$

with base-two entropies. Its bounds are $0 \le \chi_{\mathrm{P}} \le 2$ bits. One bit is the classical one-qubit benchmark; values above one bit indicate an advantage for the stated ideal protocol.

This is the dense-coding capacity when the protocol is restricted to the specified Pauli encoding with ideal transmission and asymptotic joint decoding. It is not automatically a general optimization over all encodings, a one-shot accessible-information result, or an experimentally achieved communication rate.

10.2 SOC and basis dependence

SOC enters χ_{P} through ρ_{AB} and ρ_B. There is no universal rule that increasing SOC must increase or decrease dense coding. It can change both the total entropy and the reduced entropy.

The identity Eq. (33) is valid in either qubit basis. However, original-spin and helicity-band states need not be related by fixed local unitaries, so their numerical χ_{P} values need not agree. At $\mathbf{r} = (1, 0)$,

$$\chi_{\mathrm{P}}^{\mathrm{spin}}(V_{\mathrm{SO}} = 0) = 0.000007821, \qquad \chi_{\mathrm{P}}^{\mathrm{band}}(V_{\mathrm{SO}} = 0) = 0.000003142, \tag{34}$$

whereas at $V_{\mathrm{SO}} = 0.5t$,

$$\chi_{\mathrm{P}}^{\mathrm{spin}} = 0.000013593, \qquad \chi_{\mathrm{P}}^{\mathrm{band}} = 0.002351021. \tag{35}$$

The band value is larger in this particular example, but remains far below one bit. This is not evidence of a general SOC enhancement or an operational advantage.

Useful limiting checks are

$$\rho = |\Psi^-\rangle\langle\Psi^-| \Rightarrow \chi_{\mathrm{P}} = 2, \qquad \rho = |00\rangle\langle 00| \Rightarrow \chi_{\mathrm{P}} = 1, \qquad \rho = I_4/4 \Rightarrow \chi_{\mathrm{P}} = 0. \tag{36}$$

Thus the one-bit value of a pure product state is the classical benchmark, not a residual dense-coding advantage. Concurrence, teleportation fidelity, and Holevo information must be evaluated separately.

No new noisy dense-coding dynamics are generated in this report. If a spin or band-pseudospin channel is applied, the evolved state and its reduced entropy must be recomputed with the corresponding Kraus operators. The legacy original-spin amplitude-damping results must not be relabeled as helicity-band dynamics.

11 Numerical demonstration and validation

11.1 Parameters and solver checks

The demonstration independently solves the self-consistency equations at $V_{\mathrm{SO}}/t = 0$ and 0.5 with

Parameter	Value
U/t	75
Δ_t/t	4.5
Filling n	1.875
Temperature	$T = 0$
Momentum mesh	101×101, inversion symmetric
Lattice vectors	$(0,0), (1,0), (1,1), (2,0), (0,2), (3,1)$
Bases	Original spin and helicity band

The odd mesh includes paired $\mathbf{k}, -\mathbf{k}$ points, allowing exact inversion-cancellation checks. The calculation is not a production mesh-convergence or seed-stability study.

Table 1: Executed validation diagnostics. All checks passed.

Diagnostic	Maximum absolute value
Zero-SOC solver residual	$2.906e-12$
Nonzero-SOC solver residual	$3.354e-12$
Zero-SOC Oh–Kim matrix error	$5.551e-17$
Trace error	$1.110e-16$
Hermiticity error	$0.000e+00$
Same-site singlet error	$1.110e-16$
Off-TRIM anomalous inversion residual	$3.297e-15$
Pauli-average identity error	$2.776e-17$
Holevo repository cross-check	$2.220e-16$
Bell/Werner metric error	$2.665e-15$
Minimum density-matrix eigenvalue	$-1.564e-32$

11.2 Explicit nonzero-SOC matrices

The following matrices are generated directly from the validated NPZ file. The code retains full complex values; the displayed entries are real to the shown precision.

$$\rho_{\mathrm{spin}}(V_{\mathrm{SO}} = 0.5t, \mathbf{r} = (1,0)) \approx \begin{pmatrix} 0.24961259 & -0.00049305 & 0.00049305 & 0.00023973 \\ -0.00049305 & 0.25038741 & -0.00101455 & -0.00049305 \\ 0.00049305 & -0.00101455 & 0.25038741 & 0.00049305 \\ 0.00023973 & -0.00049305 & 0.00049305 & 0.24961259 \end{pmatrix}, \quad (37)$$

$$\rho_{\mathrm{band}}(V_{\mathrm{SO}} = 0.5t, \mathbf{r} = (1,0)) \approx \begin{pmatrix} 0.27889114 & 0.00000000 & 0.00000000 & 0.00008567 \\ 0.00000000 & 0.24963159 & 0.00002221 & 0.00000000 \\ 0.00000000 & 0.00002221 & 0.24963159 & 0.00000000 \\ 0.00008567 & 0.00000000 & 0.00000000 & 0.22184567 \end{pmatrix}. \quad (38)$$

The original-spin matrix contains elements outside the Oh–Kim X pattern, including $|00\rangle\langle 11|$ coherence and local-spin off-diagonal structure. The band matrix has a different diagonal and coherence structure. Forcing either matrix into the scalar Werner form would discard information.

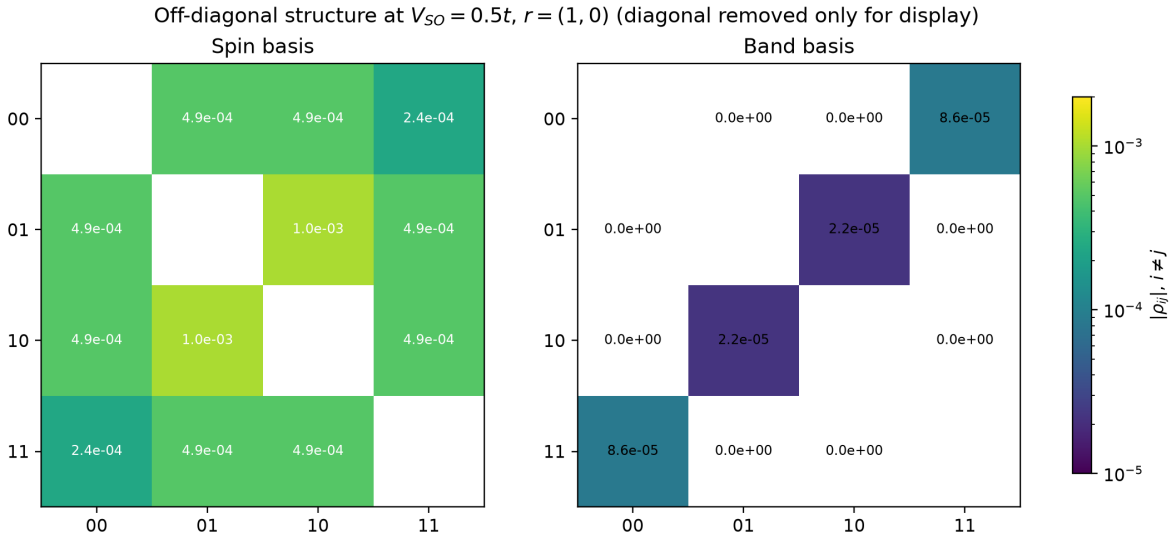


Figure 1: Off-diagonal magnitudes for the two matrices above. The diagonal is removed only for visualization. Original-spin and band-pseudospin states have different matrix structures and are not interchangeable.

11.3 Resource table

Table 2: Validated $V_{\mathrm{SO}} = 0.5t$ resources at exact lattice vectors. $\overline{\mathcal{F}}_{\mathrm{ref}}$ is fixed-reference teleportation, $\overline{\mathcal{F}}_{\mathrm{opt}}$ is locally optimized average teleportation, and χ_{P} is the implemented Pauli-ensemble Holevo quantity. No state below the top singlets has an operational advantage.

Basis	**r**	C	$\overline{\mathcal{F}}_{\mathrm{ref}}$	$\overline{\mathcal{F}}_{\mathrm{opt}}$	χ_{P}	$S_2(\rho_B)$
Spin	$(0,0)$	1.000000000	1.000000000	1.000000000	2.000000000	1.000000000
Spin	$(1,0)$	0.000000000	0.500934644	0.501254157	0.000013593	1.000000000
Spin	$(1,1)$	0.000000000	0.500576534	0.500938408	0.000007613	1.000000000
Spin	$(2,0)$	0.000000000	0.500022720	0.500371458	0.000001194	1.000000000
Spin	$(0,2)$	0.000000000	0.500022720	0.500371458	0.000001194	1.000000000
Spin	$(3,1)$	0.000000000	0.499980198	0.500077918	0.000000053	1.000000000
Band	$(0,0)$	1.000000000	1.000000000	1.000000000	2.000000000	1.000000000
Band	$(1,0)$	0.000000000	0.499739591	0.500302714	0.002351021	0.997651326
Band	$(1,1)$	0.000000000	0.499758658	0.500492092	0.002285567	0.997715465
Band	$(2,0)$	0.000000000	0.499455102	0.500623183	0.002201482	0.997798638
Band	$(0,2)$	0.000000000	0.499455102	0.500623183	0.002201482	0.997798638
Band	$(3,1)$	0.000000000	0.499486978	0.500525833	0.002207541	0.997792461

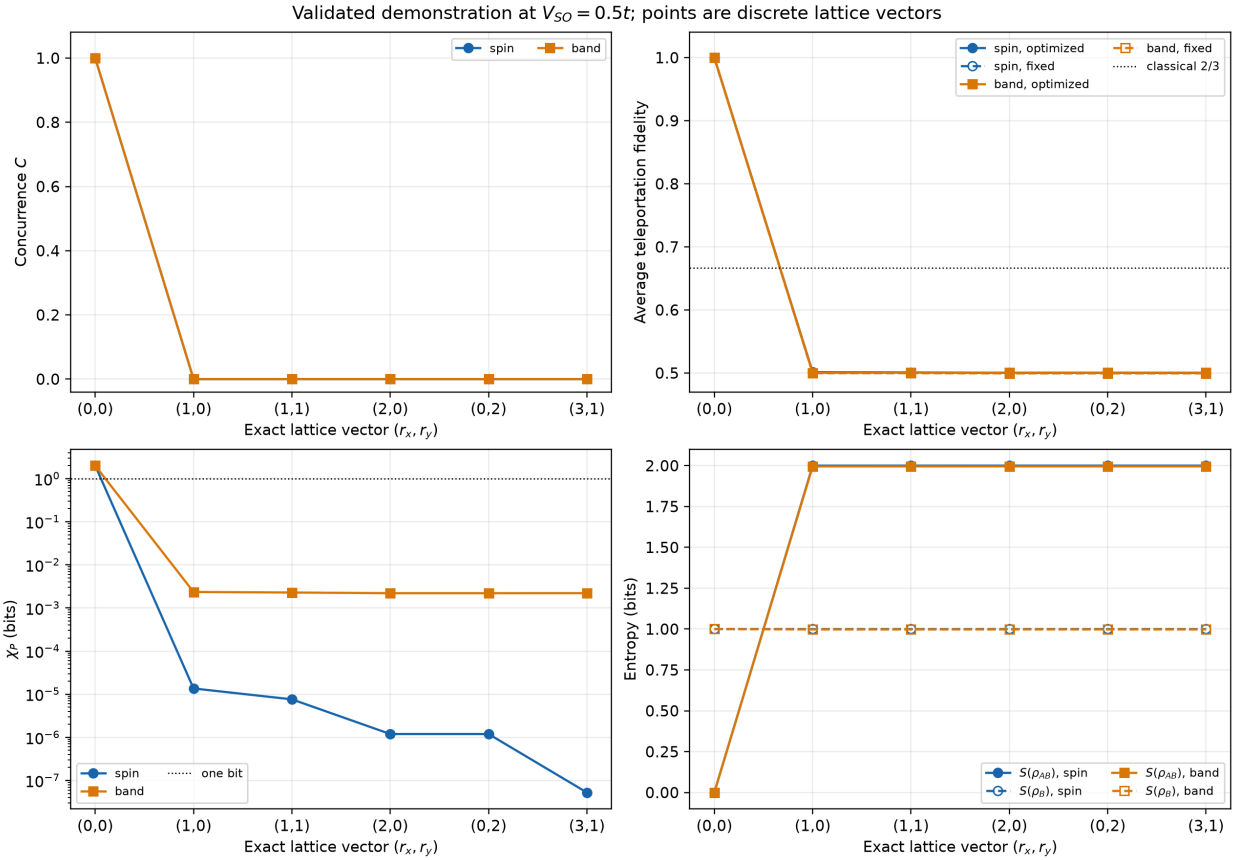


Figure 2: Discrete-lattice resource summary at $V_{\mathrm{SO}} = 0.5t$. Lines connect discrete points only as visual guides. The same-site singlet has ideal values; all tested nonzero-site states remain below the teleportation and dense-coding benchmarks.

12 Algorithm and code map

The implemented algorithm is:

1. Solve the BdG self-consistency equations for the chosen $U, \Delta_t, n, V_{\mathrm{SO}}, N_k, T$.
2. Construct $v^2_{\mathbf{k}s}$ and $\bar{\Delta}_{\mathbf{k}s}/(2E_{\mathbf{k}s})$.
3. Restore the full phase $se^{-i\theta_{\mathbf{k}}}$ and enforce TRIM/Pauli constraints.
4. Build matrix-valued band correlators using Eqs. (7)–(8).
5. For original spin, apply Eq. (10); for band qubits, retain the band matrices.
6. Fourier-transform to A, B using Eq. (11).
7. Evaluate all 16 elements with the Wick formula Eq. (13).
8. Retain P_{AB}, normalize once, and check trace, Hermiticity, and positivity.
9. Calculate general concurrence, fixed-reference fidelity, FEF, optimized teleportation fidelity, entropies, and χ_{P} separately.

```
from helicity_density_matrix import (
    density_at_separation, resource_metrics, dense_coding_metrics)

spin = density_at_separation(bdg, [1, 0], basis="spin")
band = density_at_separation(bdg, [1, 0], basis="band")

print(spin["weight"], resource_metrics(spin["rho"]))
print(dense_coding_metrics(spin["rho"]))
print(band["weight"], resource_metrics(band["rho"]))
print(dense_coding_metrics(band["rho"]))
```

The main files are:

- `helicity_density_matrix.py`: correlators, Wick construction, general concurrence, teleportation, and dense-coding metrics;
- `analysis/validate_helicity_density_matrix.py`: fresh solver runs and validation;
- `data/helicity_density_matrix_demo.npz`: matrices, correlators, weights, and observables;
- `reports/helicity_density_matrix_validation.json`: complete diagnostics;
- `analysis/generate_helicity_report_assets.py`: report tables and figures generated from the NPZ.

13 Scope, limitations, and next calculations

1. **Gaussian mean-field scope.** Wick factorization applies to the quadratic BdG state, not automatically to an exact non-Gaussian Hubbard ground state.
2. **Pairing scope.** The formulas assume spatially uniform zero-center-of-mass-momentum intraband pairing. Interband pairing, FFLO states, disorder, or position-dependent gaps require their full correlators.
3. **Conditional detection.** The result is not a canonical fixed-total-number reduction of the full BCS state.
4. **Numerical scope.** Two SOC values and one mesh do not establish a continuous SOC dependence. Production work requires seed and mesh convergence, occupation checks, and independent exact-site evaluation.
5. **Operational scope.** Band-pseudospin entanglement is not evidence that helicity Bell measurement, local Pauli feedforward, or dense-coding transmission is experimentally available.
6. **Dynamical scope.** No new environmental evolution is asserted. Noise acts on a specified qubit basis and must be applied to that basis's density matrix.
7. **No automatic positive result.** A larger gap or anomalous correlator is not, by itself, evidence of teleportation or dense-coding advantage.

The next scientifically useful calculation is a production V_{SO} sweep with independently reconverged BdG solutions, at least two momentum meshes, exact lattice-site resources, and separate original-spin and band-pseudospin labels. Only then should trends with SOC be discussed.

A Common sign and convention errors

- **Dropping $se^{-i\theta_{\mathbf{k}}}$.** This confuses $\bar{\Delta}_{\mathbf{k}s}$ with the full gap and violates the required pair structure.
- **Using $UNU^\dagger$ blindly.** For $N_{\sigma\sigma'} = \langle c^\dagger_\sigma c_{\sigma'}\rangle$, the implemented spin transformation is $U^* N U^T$.
- **Using $U_{\mathbf{k}}$ for both anomalous operators.** The pair has momenta $\mathbf{k}, -\mathbf{k}$, so the transformation contains $U_{\mathbf{k}} F_{\mathbf{k}} U^T_{-\mathbf{k}}$.
- **Ignoring TRIM points.** Nondegenerate same-band pairing vanishes where the two time-reversed operators coincide.
- **Using F^*_{BA} with the wrong sign.** The creation-side factor in the Wick matrix is $F^*_{AB,\alpha'\beta'} = -F^*_{BA,\beta'\alpha'}$.
- **Forgetting same-site antisymmetry.** $F_{AB}(0)$ must be antisymmetric; finite-grid residue is not a physical same-spin onsite pair.
- **Projecting onto an X state.** This hides precisely the new SOC-generated matrix elements.
- **Clipping negative eigenvalues.** Positivity must follow from the physical contraction; clipping would conceal implementation errors.
- **Equating spin and band protocols.** The Pauli algebra is identical, but the physical modes and required operations differ.

B Reproducibility

From the repository root:

```
.venv/bin/python analysis/validate_helicity_density_matrix.py
.venv/bin/python analysis/generate_helicity_report_assets.py
```

The validation writes only helicity-specific artifacts and does not overwrite archived $V_{\mathrm{SO}} = 0$ Green's-function files. The report assets script reads the validated NPZ and generates every numerical table and both report figures. The PDF is compiled independently of `manuscript.pdf`.

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